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Optimizing Heterogeneous High-Performance Computing Resource Allocation for Optimal Performance in Advanced Seismic Imaging Algorithms

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Abstract

Optimizing resource allocation for seismic imaging workloads in High-Performance Computing (HPC) environments, particularly in heterogeneous infrastructures with diverse architectures, is crucial for achieving optimal computational efficiency in the oil and gas industry. As seismic imaging algorithms continue to grow in sophistication, they exhibit diverse resource requirements, making it essential to understand and accommodate these varying demands to ensure optimal resource utilization. This paper presents a data-driven resource allocation strategy that leverages application-level behavior metrics to optimize workload placement and resource utilization in heterogeneous HPC environments. The methodology is centered around defining two key metrics: Application CPU Ratio (ACPR) and Size CPU Ratio (SCR), which enable accurate identification of the applications' resource consumption patterns to guide decisions on optimal resource allocations. By identifying the optimal number of cores required for each application, this approach maximizes system throughput and eliminates idle resources. The proposed optimization method has been successfully applied to production HPC environment, without requiring any modifications to the underlying infrastructure, resulting in significant improvements in system efficiency, with a substantial 100X decrease in queue wait times, enhancing overall resource utilization. The methodology's effectiveness underscores its potential to transform resource management in various scientific and engineering domains.

Introduction

High-Performance Computing (HPC) has become a foundational driver of digital transformation in the oil and gas industry, particularly in seismic imaging and reservoir characterization workflows. These workflows require significant computational power and memory resources due to the scale and complexity of modern subsurface datasets (Al Dulaijan and Alajaji, 2022). As seismic imaging algorithms continue to evolve, their demands on computing infrastructure grow, not only in volume, but in variability.

Unlike conventional HPC domains that typically involve uniform computational patterns, seismic imaging consists of multiple algorithmic stages with heterogeneous performance requirements. For instance, an imaging project may simultaneously execute Surface-Related Multiple Elimination (SRME), Kirchhoff

Migrations, Full Waveform Inversion (FWI), and Tomography, each exhibiting distinct resource usage behaviors. Some algorithms are compute-bound, while others are constrained by memory bandwidth or capacity.

In many oil and gas organizations, these workflows run on heterogeneous HPC environments that include clusters with varied CPU architectures, memory configurations, and interconnect topologies. Managing workload distribution across such diversity remains a significant challenge. Conventional job scheduling systems assume functional uniformity across compute nodes and do not account for the specific resource needs of each application. This complicates workload placement and often results in inefficient resource allocation, even when aggregate system capacity is sufficient.

To address this challenge, a data-driven scheduling strategy has been developed. It establishes two quantitative metrics—Application CPU Ratio (ACPR) and Size CPU Ratio (SCR)—allowing for accurate profiling of application resource consumption to guide resource allocation decisions. The method was developed and deployed in a production HPC environment supporting seismic workloads, where it demonstrated significant gains in system efficiency and workload throughput.

Background

Seismic imaging workflows in the oil and gas industry are resource-intensive and computationally diverse. Applications such as Surface-Related Multiple Elimination (SRME), Kirchhoff Migrations, and Tomography, present distinct and contrasting computational demands. SRME and Kirchhoff Migrations in general are highly compute-intensive processes, requiring vast parallel processing power and fast random access to data to handle its intensive mathematical calculations. In contrast, Tomography is primarily memory-bound, demanding significant memory capacity and exhibiting sequential access patterns during its inversion process, characterized by large-scale matrix operations and dense linear algebra computations (Al-Shaikh et al., 2012). These variations reflect the broader spectrum of workload behavior present across full-scale seismic processing pipelines.

This diversity introduces substantial complexity into resource management within HPC environments. Imaging workflows comprise algorithmic stages with divergent performance demands; some are constrained by CPU cycles, while others are limited by memory or I/O throughput. Such variability complicates workload placement and often results in inefficient resource allocation, even when total system capacity is not fully utilized. In such environments, core allocation strategies that fail to consider application-specific behavior often lead to inefficient provisioning. Applications may be assigned too few or too many cores relative to their actual needs, resulting in either wasted resources or system congestion. Moreover, these workloads often execute within shared HPC infrastructures supporting multiple users and a variety of scientific applications. The heterogeneous computational demands lead to resource contention, as applications vary in their usage of compute nodes, memory bandwidth, and network interconnects. This variability can result in uneven resource utilization and fragmentation, thereby reducing overall system efficiency.

Workloads are often directed to specific clusters with oversubscribed resources, leading to underutilization and leaving valuable compute power idle, while many jobs experience increased queue wait times. This highlights the need for a balanced system; one that maximizes throughput and resource potential while minimizing wait times and idle capacity. Such a system must consider the diverse and competing demands placed on HPC infrastructure by varying seismic imaging applications, ensuring that resources are allocated in a manner that aligns with actual workload requirements and system capabilities.

Effective resource management in this context requires not only awareness of application-specific behavior but also a comprehensive view of cluster-wide utilization patterns. By balancing these perspectives, scheduling policies can mitigate resource fragmentation and prevent bottlenecks that commonly arise in heterogeneous HPC environments.

To address these challenges, a behavior-aware scheduling strategy is required, one that systematically leverages measured per-application profiling data to guide and optimize resource allocation decisions across the HPC environment. Through this approach, resource allocation decisions are grounded in accurate workload characterization, facilitating improved system efficiency and throughput.

Methodology

This section presents the core resource allocation strategy developed to optimize workload scheduling in heterogeneous HPC environments supporting seismic imaging workflows. The method is based on data-driven application profiling and the derivation of two key quantitative metrics to guide proportional core allocations. HPC environment heterogeneity introduces substantial complexity into resource management within HPC environments. Imaging workflows are composed of algorithmic stages with varying performance demands; some are constrained by CPU cycles, while others are constrained by memory or I/O throughput. This variation complicates workload placement and often results in inefficient resource allocation, even when total system capacity is not fully utilized.

To construct this model, we analyzed historical workload execution data using Python-based analytics tools (e.g., pandas). Each workload, corresponding to a submitted seismic imaging job, was profiled to extract key resource utilization indicators, including CPU time (seconds), memory usage (GB), and node elapsed time (seconds). Applications were grouped and aggregated to identify high-impact workloads, defined as those contributing most significantly to overall system consumption. For each selected application, we computed the cumulative CPU time across all jobs to establish a normalized usage ratio. This profiling process enabled the identification of highly utilized and resource-intensive applications, serving as a foundation for defining quantitative indicators that inform proportional core allocation and workload placement.

Profiling and Metric Definition

Two key quantitative metrics were developed to accurately capture application-specific resource behavior on a per-application basis: Application CPU Ratio ($ACPR_i$) and Size CPU Ratio (SCR_i). These metrics serve distinct but complementary roles in quantifying how efficiently an application utilizes its allocated cores and memory resources, respectively. By measuring both computational concurrency and memory intensity, the metrics collectively provide a comprehensive characterization of workload demands. This dual-metric approach enables precise proportional core allocation and node selection decisions, thereby optimizing workload placement and resource utilization tailored to the unique execution profiles of individual applications within complex seismic imaging workflows.

Eq. 1 defines the Application CPU Ratio for application i :

$$ACPR_i = \frac{\sum Job \ CPU \ Time_i}{\sum Application \ CPU \ Time \ i} \quad (1)$$

Where:

$\sum Job \ CPU \ Time_i$ denotes the total CPU time reported for each job instance containing application i , and $\sum Application \ CPU \ Time_i$ refers to the cumulative CPU time attributable specifically to application i , aggregated across all job instances where it was executed; if jobs include multiple applications, CPU time is partitioned accordingly.

Eq. 2 defines the Size CPU Ratio for application i :

$$SCR_i = \frac{\max_{j=1}^N Application \ CPU \ Time_j}{\sum Application \ CPU \ Time_i} \quad (2)$$

Where:

N is the total number of applications,

Max Application CPU Time is the maximum CPU time consumed by any application j among all N applications and $\sum Application\ CPU\ Time_i$ refers to the cumulative CPU time attributable specifically to application i , aggregated across all job instances where it was executed; if jobs include multiple applications, CPU time is partitioned accordingly.

Applications are selected for optimization if both $ACPR_i < T_1$ and $SCR_i < T_2$. T_1 and T_2 are threshold parameters, typically set to 1 and 10, respectively.

The selection process is thus performed individually for each application, allowing granular identification of high-impact workloads.

Core Allocation Model

The profiling metrics enable a proportional core allocation model that aligns the number of optimal cores requested with the application's observed execution characteristics, primarily derived from measured CPU consumption and node elapsed time.

- **Step 1:** Calculate $ACPR_i$ and SCR_i for each application as described in Section 3.1.
- **Step 2:** Determine the number of optimal cores per application using Eq. 3:

$$Cores_i = \frac{\sum Node\ Elapsed\ Time\ i}{\sum Application\ CPU\ Time\ i} \quad (3)$$

Where:

$\sum Node\ Elapsed\ Time\ i$ is the total elapsed wall-clock time that nodes are occupied by application i , and $\sum Application\ CPU\ Time\ i$ refers to the cumulative CPU time attributable specifically to application i , aggregated across all job instances where it was executed; if jobs include multiple applications, CPU time is partitioned accordingly.

- **Step 3:** Calculate the overall number of optimal cores for the submission environment as the average over the N selected applications, as given by Eq. 4:

$$Optimal\ Cores = \frac{\sum_{i=1}^N Cores_i}{N} \quad (4)$$

Where:

N is the number of applications selected for optimization based on the metrics in Section 3.1. This average reflects the recommended core allocation that balances efficiency and workload requirements across the relevant applications.

This multi-step approach provides a robust mechanism to balance core allocation by reflecting both individual application behavior and overall workload composition, improving accuracy over heuristic-based methods.

Data quality and profiling accuracy are critical factors influencing the reliability of this model. Therefore, the profiling process incorporates data cleansing steps to exclude incomplete or anomalous records, and applies statistical smoothing techniques to mitigate the impact of outliers. These measures ensure that the derived metrics accurately reflect typical application performance, avoiding skewed core allocations that could degrade scheduling efficiency.

Node Selection and Scheduling Integration

Using the calculated number of optimal cores, the scheduler selects nodes with available resources that best match the scaled resource allocation of selected applications, considering heterogeneous CPU architectures and memory configurations. The method integrates directly into the job submission process, offering explicit core allocation recommendations based on measured profiling rather than static heuristics. This integration improves workload placement accuracy and reduces the risk of overallocation.

Importantly, the approach requires no modifications to existing hardware configurations or underlying scheduling algorithms. Instead, it refines job submission parameters, specifically the core count requests, enabling the existing scheduler to place workloads more effectively based on actual resource demands. By providing core count guidance derived from dynamic application profiling, the system enhances utilization across heterogeneous nodes and reduces resource fragmentation and idle compute capacity. This results in more precise resource allocation within the scheduler's current operational framework.

The profiling data can be periodically updated to capture evolving application behaviors, ensuring that core allocation recommendations remain aligned with real-world workload behavior. This adaptive mechanism helps maintain scheduling efficiency over time in production HPC environments.

Integration into the existing submission pipeline is achieved through adjustments to job resource requests at submission time, requiring no changes to scheduling policies, algorithms, or system software. This scheduler-agnostic integration ensures compatibility with existing HPC workflows while delivering measurable improvements in throughput and resource utilization.

Results and Evaluation

The optimization methodology was applied in a production-scale heterogeneous HPC environment supporting seismic imaging workloads. The production HPC infrastructure supports a wide range of memory-bound and compute-intensive applications, each exhibiting distinct concurrency profiles and resource consumption patterns.

Prior to implementation, core allocation was based on established provisioning practices that ensured broad applicability across workloads, and was further enhanced through the integration of workload-specific concurrency profiling. To evaluate the effectiveness of this method, two key operational performance indicators were examined: Queue Wait Time and System Throughput. Together, these metrics characterize the scheduler's capability to maximize resource utilization and improve workload placement. All reported results are derived from production job submissions executed under standard operational workflows and conditions.

The application-level profiling metrics provided key insights that contributed to optimizing overall resource allocation, supporting significant reductions in queue wait times, eliminating overallocations, and contributing to substantial improvements in throughput.

Queue Wait Time Optimization

Under previous configurations, average job wait times corresponded to approximately 700 hours, reflecting resource overallocation and the scheduling complexity inherent in a high-performance, shared production environment. By integrating core counts calculated through the profiling-based estimation method into the job scheduling process, wait times decreased significantly to approximately 7 hours, resulting in a 100X reduction. This led to enhanced resource utilization and scheduling efficiency.

More detailed analysis reveals that the reduction in queue wait times was particularly pronounced for medium-to-large scale jobs, which traditionally faced longer delays due to their resource demands. Smaller jobs also benefited from this optimization, experiencing improved turnaround times as the scheduler more accurately matched core requests to actual application requirements. This improvement translated directly to enhanced operational efficiency, reducing idle times for compute nodes and accelerating project timelines.

The observed improvement is attributed to increased accuracy in resource allocation. Core requests were adjusted based on quantified concurrency profiles of each application, enabling more effective workload distribution across compute nodes. These enhancements were achieved without modifications to hardware infrastructure or scheduling policies, demonstrating the efficacy of the profiling-based allocation method.

The production environment's queue wait times decreased significantly after integrating our optimization methodology, as shown in the graph in Fig. 1. This improvement demonstrates the environment's responsiveness to optimized core allocation strategies.

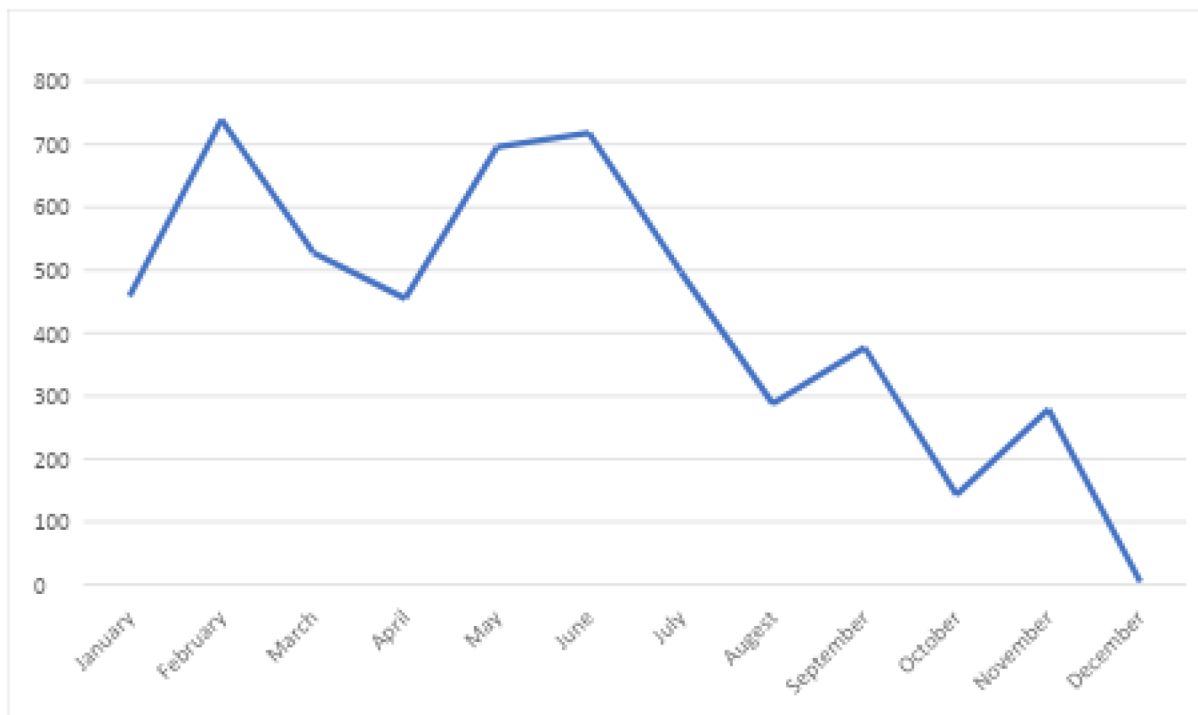


Figure 1—presents the measured queue wait times across the production environment over time. The sharp decline observed during the final month corresponds with the integration of the optimization methodology. This visual trend confirms the system's responsiveness to improved core allocation inputs, under consistent hardware and scheduling conditions.

The underlying infrastructure remained unchanged, confirming that performance gains resulted from refined scheduling inputs rather than increased capacity.

Throughput Enhancement

In addition to reducing wait times, the method also contributed to a substantial increase in overall workload throughput. By aligning core requests with concurrency profiles derived from observed workload execution data, the scheduler accommodated a larger number of jobs simultaneously without overcommitting system resources. This capability supported a higher volume of processing completed within the same scheduling interval.

Additional metrics, such as job completion rate and average job turnaround time, were observed to improve correspondingly. This indicates that the optimized core allocation not only increased the number of jobs processed but also contributed to faster job execution cycles, enhancing overall system responsiveness. These throughput gains align closely with business objectives to accelerate seismic data processing and reduce project turnaround times.

This increase in throughput validates the method's capability to enhance existing scheduling mechanisms. Implementation was achieved through adjustments to job submission parameters alone,

without necessitating changes to existing scheduling policies or the underlying system architecture, thereby enabling seamless integration into operational workflows.

Resource Utilization Efficiency

Beyond wait time and throughput improvements, the method enhanced CPU utilization efficiency, defined as the ratio of consumed CPU time to reserved node time. In scenarios where application scaling behavior was non-linear, static core provisioning could lead to underutilized compute capacity. The introduced optimization method provided a means to identify and mitigate such inefficiencies.

As a result, across the evaluated dataset, average CPU utilization per node increased consequently supporting a closer alignment between allocated and consumed compute time. This refinement has strengthened the system's ability to accommodate heterogeneous workloads without compromising throughput or responsiveness.

Practically, improved CPU utilization translates to cost savings and environmental benefits through reduced energy consumption per unit of computation. This optimization also increases the effective capacity of existing HPC resources, delaying or potentially reducing the need for costly hardware expansions.

Discussion

The results demonstrate that integrating workload-specific concurrency profiling into core allocation significantly enhances HPC system efficiency. By aligning resource requests with precise application demands, the approach effectively addresses the heterogeneity inherent in seismic imaging workflows without requiring modifications to existing scheduling infrastructures. This highlights the method's practical applicability within production environments where system stability and operational reliability are critical.

The substantial reduction in queue wait times, coupled with increased throughput, evidences more effective utilization of available compute resources. These improvements indicate that while conventional static provisioning strategies are broadly effective, they can benefit from refinement through data-driven resource characterization. Notably, the optimization respects operational constraints by leveraging historical profiling data to inform resource allocation decisions that balance system load and application performance. Furthermore, the observed enhancement in CPU utilization efficiency highlights the importance of tailoring core requests to the non-linear scaling behaviors exhibited by various seismic imaging applications. By mitigating overallocation, the method reduces idle compute capacity and increases the scheduler's capacity to accommodate a greater volume of heterogeneous workloads concurrently.

The implications of this approach extend beyond seismic imaging, offering valuable insights for HPC resource management in other scientific and engineering domains characterized by heterogeneous and dynamic workload demands. The presented methodology constitutes a scalable framework adaptable to diverse HPC contexts where comprehensive workload characterization is attainable. Although this study focuses on batch job submissions informed by historical profiling data, there is significant potential to advance this approach through the incorporation of real-time monitoring and dynamic scheduling techniques. The integration of live profiling data may facilitate more responsive resource allocation, adapting to transient workload variations and thereby further optimizing system throughput and minimizing latency. Implementing such adaptive scheduling strategies poses challenges related to profiling overhead, data precision, and compatibility with existing HPC resource management systems. Nevertheless, ongoing advancements in HPC monitoring technologies and scheduling frameworks are progressively addressing these limitations, enhancing the feasibility of dynamic, behavior-aware scheduling.

In summary, the findings of this study advocate for continued investigation into hybrid resource allocation models that combine data-driven profiling with predictive analytics and machine learning methodologies. Such multifaceted approaches have the potential to improve the accuracy and responsiveness of resource

allocation decisions, thereby advancing HPC system performance within increasingly complex and diverse computational environments.

Conclusion

This paper presented a data-driven method for optimizing core allocation in heterogeneous HPC environments supporting seismic imaging workflows. By introducing quantitative metrics derived from empirical concurrency profiling, the approach enables precise alignment of resource requests with application-specific demands. Its deployment within a production environment yielded significant performance improvements, including a reduction in queue wait times from approximately 700 hours to 7 hours alongside a substantial increase in workload throughput and CPU utilization efficiency.

These results demonstrate that workload-aware scheduling strategies, grounded in measured application behavior, can substantially elevate system performance while achieving considerable resource savings. The method's ability to improve throughput and resource utilization without requiring modifications to existing scheduling policies or hardware infrastructure demonstrating its practical value and operational scalability.

The findings emphasize the importance of integrating detailed workload characterization into HPC resource management frameworks, particularly in domains characterized by heterogeneous and dynamic computational demands. By tailoring resource allocation to actual application behavior, this approach contributes to more efficient and sustainable HPC operations.

Looking forward, the methodology provides a foundation for further advancements in adaptive and predictive resource scheduling, with potential applications across a broad range of scientific and engineering disciplines. The integration of real-time profiling and advanced analytics presents promising avenues for enhancing responsiveness and optimization in future HPC systems.

References

- Al Dulaijan, K., and Alajaji, M. [2022]. HPC advances and impacts to exploration and reservoir development. SEG/AAPG International Meeting for Applied Geoscience & Energy.
- Al-Shaikh, R. A., Abulleif, S., and Luo, Y. [2012]. Seismic attribute analysis for reservoir characterization. *Search and Discovery Article #90141Geo*.